

COKE_OVEN_PPE_SOP

STEEL AUTHORITY OF INDIA LIMITED
BOKARO STEEL PLANT

Standard Operating Procedure

SOP No: BSP-CO-007

Title: Personal Protective Equipment Requirements — Coke Oven Battery

Department: Coke Ovens

Revision: 01

Date: 10-Feb-2024

1. PURPOSE

This SOP defines the mandatory Personal Protective Equipment (PPE) requirements for all personnel working in or entering the Coke Oven Battery area at Bokaro Steel Plant.

2. SCOPE

Applicable to all coke oven operators, maintenance staff, supervisors, safety officers, and visitors entering the following areas:

- Coke Oven Battery Nos. 1 through 6
- Coal Tower and Coal Handling areas
- Coke Sorting Plant
- By-Product Plant
- Pusher Car and Quenching Car operation zones

3. HAZARD IDENTIFICATION — COKE OVEN AREA

- High temperature surfaces (oven doors, standpipes): up to 1200°C
- Coke oven gas (COG): toxic and flammable
- Coal dust: respiratory hazard, explosive above 35 g/m³
- Benzene vapours: carcinogenic, present in by-product plant
- Hot coke emissions during pushing operations
- UV and IR radiation during door opening

4. MANDATORY PPE — ALL ZONES

4.1 Head Protection

- Hard hat (IS 2925 compliant, minimum Class A)
- Must be inspected before each shift for cracks or damage
- Replace immediately if any impact has occurred

4.2 Eye and Face Protection

- Safety goggles (chemical splash resistant) — mandatory at all times
- Additional face shield with IR filter (shade 3-5) required during:

- Oven door opening
- Hot coke pushing operations
- Any work within 5 metres of standpipes

4.3 Respiratory Protection

- Half-face respirator with P100 + OV cartridge: mandatory in battery area
- Full-face SCBA: mandatory during emergency response or confined space entry
- Disposable dust mask (FFP3): NOT acceptable as substitute in active battery area
- Filter cartridges must be replaced every 8 hours or immediately if breathing resistance increases

4.4 Hand Protection

- Heat-resistant leather gauntlets: mandatory during door operations
- Chemical-resistant nitrile gloves: mandatory in by-product plant
- Do NOT use cotton gloves — they provide no protection against hot surfaces

4.5 Body Protection

- Flame-retardant coveralls (FR rating: minimum ATPV 8 cal/cm²)
- High-visibility vest over coveralls at all times
- Coveralls must not have exposed metal buttons or zippers near oven fronts

4.6 Foot Protection

- Safety boots with steel toe cap and heat-resistant sole (minimum 300°C)
- Metatarsal guard required for pusher car and coke guide operators
- No synthetic-material footwear permitted in the battery area

4.7 Hearing Protection

- Ear plugs (SNR 28 dB minimum) mandatory in all battery zones
- Noise levels in battery area: 90-105 dB — exceeds permissible limit
- Ear plugs must be fitted correctly — consult safety officer for fitting

5. ADDITIONAL PPE — SPECIFIC TASKS

5.1 Oven Door Removal and Replacement

- Full aluminized heat-reflective suit required
- Face shield with shade 5 IR filter
- Leather gauntlet gloves over base gloves

5.2 Quenching Car Operations

- Waterproof outer layer over FR coveralls
- Additional rubber boots over safety boots

5.3 Confined Space Entry (e.g. collector main, gas offtake)

- Full SCBA (not air-purifying respirator)
- Safety harness with lifeline
- Buddy system mandatory — minimum 2 persons

6. PPE INSPECTION AND MAINTENANCE

- All PPE must be inspected before each shift
- Damaged PPE must be reported to supervisor and replaced immediately
- PPE is personal issue — do not share respiratory equipment
- Cleaning schedule: coveralls washed weekly, goggles cleaned daily

7. VISITOR REQUIREMENTS

All visitors must be provided with and must wear:

Hard hat, safety goggles, FR coverall, safety boots, ear plugs

No visitor may enter active battery area without a trained escort.

8. NON-COMPLIANCE

Failure to wear mandatory PPE is a serious safety violation.

First instance: written warning

Second instance: suspension from site access pending safety retraining

Third instance: permanent removal from site